

Question:

If the sum of all the angles except one of a convex polygon is 2190° , then the number of sides of the polygon must be

- (A) 13 (B) 15 (C) 17 (D) 19 (E) 21

Reasoning + Answer:

<think>

1. Understand the problem:

- We are given that the sum of all the angles except one of a convex polygon is 2190° .
- We need to find the number of sides (n) of the polygon.

2. Use the formula for the sum of interior angles of a polygon:

- The sum of the interior angles of an n -sided polygon is given by $(n - 2) \times 180^\circ$.

3. Set up the equation:

- Let x be the angle that is excluded from the sum.
- Therefore, the sum of the remaining $(n - 1)$ angles is 2190° .
- The sum of all the interior angles is $(n - 2) \times 180^\circ$.
- Thus, we have:

$$(n - 2) \times 180^\circ = 2190^\circ + x$$

4. Since x is part of the interior angles and the polygon is convex, the excluded angle x must be less than 180° .

5. Consider the possible values of n :

- We need to find n such that the sum of the remaining angles equals 2190° .

6. Test the given options:

- Option (A): $n = 13$
 - Sum of interior angles for 13 -sided polygon: $(13 - 2) \times 180^\circ = 19 \times 180^\circ = 3420^\circ$
 - Remaining sum: $3420^\circ - 2190^\circ = 1230^\circ$
 - This does not fit the requirement.
- Option (B): $n = 15$
 - Sum of interior angles for 15 -sided polygon: $(15 - 2) \times 180^\circ = 13 \times 180^\circ = 2340^\circ$
 - Remaining sum: $2340^\circ - 2190^\circ = 150^\circ$
 - This does not fit the requirement.
- Option (C): $n = 17$
 - Sum of interior angles for 17 -sided polygon: $(17 - 2) \times 180^\circ = 15 \times 180^\circ = 2700^\circ$
 - Remaining sum: $2700^\circ - 2190^\circ = 510^\circ$
 - This does not fit the requirement.
- Option (D): $n = 19$
 - Sum of interior angles for 19 -sided polygon: $(19 - 2) \times 180^\circ = 17 \times 180^\circ = 3060^\circ$
 - Remaining sum: $3060^\circ - 2190^\circ = 870^\circ$
 - This does not fit the requirement.
- Option (E): $n = 21$
 - Sum of interior angles for 21 -sided polygon: $(21 - 2) \times 180^\circ = 19 \times 180^\circ = 3420^\circ$
 - Remaining sum: $3420^\circ - 2190^\circ = 1230^\circ$